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Application for the Ph.D. position in the HypoWaves project — Bio 01/2026

Dear Dr Henkel, dear Dr Ahmerkamp,

Advective porewater flow is what turns permeable sand into a biocatalytic filter, and the paper Fabian Lange and Dr Ahmerkamp published this year (Lange et al. 2026) gives the organic matter in that filter a dual role. Glycan concentration scales with volumetric oxygen consumption, and the same glycans stabilise the bedforms — migration slowed by almost an order of magnitude against clean sand — which sustains the advective supply of oxygen into the bed. The same molecules feed the demand and hold open the supply. The study hypothesises bedform stability as a key control on remineralisation and fixed-nitrogen loss, and then measures the carbon and the oxygen; the nitrogen rates are still unmeasured. That is the measurement I would like to make.

The pelagic half of the same question is open in your own words. In the 2025 coastal review you write that “*the conditions under which nitrification or denitrification becomes the most important N_2O production pathway in coastal areas, especially in the short term, have yet to be identified*” (Voss et al. 2025). Wan and colleagues (2023), whom that review cites, showed that even in well-oxygenated coastal water the reductive pathway supplies most of the N_2O and collapses when the particles are filtered out — but the particle enters their analysis only as bulk quantity, suspended matter and particulate nitrogen: no organic composition, no lability, nothing about what the carbon is. So the question I would want to test is whether what a Baltic aggregate holds and releases as N_2O is set by how fast oxygen falls and returns rather than by how low it goes, and whether the aggregate’s own organic matter decides which. Beside the ^{15}N rates I would put a compound-specific carbon measurement on the particle’s own lipids. It fails cleanly: if the partitioning tracks the oxygen minimum alone, the idea is wrong and should be reported so.

The microbial half is the part I find hardest to put down. Dr Ahmerkamp, in the Janssand work with Hannah Marchant (Marchant et al. 2018) you looked for a transcriptional response across the tidal cycle, found no strong one, and concluded that community composition rather than regulation drives the imbalance in the denitrification intermediates. I have wondered since whether a single tidal day can separate the two, and whether rapid, repeated oxygen swings would be the forcing under which selection and regulation part company. HypoWaves could test that. Dr Henkel, your *Sulfurimonas* work (Henkel et al. 2022) taught me the opposite and equally necessary discipline: isolate the organism, measure the rate per cell, and only then scale it — a cell-specific rate multiplied by in-situ abundance reproducing a Black Sea profile that abiotic chemistry could not explain. That is the argument I would want to be able to make for N_2O .

Transport is the line through my own work, and crossing between disciplines to follow it is the habit I would bring. My master’s thesis ran across three institutions at once — MARUM, the Max Planck Institute for Marine Microbiology and the University of Bremen, within the DFG collaborative research centre TRR 420 — a geoscientist’s question answered in a glycobiology laboratory. In Jan-Hendrik Hehemann’s group I built a model system for sinking particles, followed their degradation by marine *Vibrio* strains image by image, and carried the result through a settling model to the exponent of the Martin curve — how much carbon survives to depth follows from how fast the particle is taken apart.

On Helgoland Mud Area sediments I designed ^{13}C -acetate incubations and followed the label into single fatty acids by GC-C-IRMS, separating bacterial from archaeal activity across the redox zonation; in the porewater programme on the same core I measured the nutrient and sulfide profiles and read the fluxes with Fick’s law after testing for steady state. Comparing that core with itself weeks later gave me, unintentionally, a record of what happens when oxygen reaches reduced sediment: sulfide oxidised away, iron and manganese re-precipitating, phosphate re-adsorbed onto the fresh oxides. HypoWaves asks that question on purpose. At the AWI organic geochemistry laboratories I ran ultrasonic extraction, open-column chromatography, silylation, and GC-MS and HPLC-MS quantification of highly branched isoprenoids and GDGTs.

The methods are the part I have been preparing for. From the GLOMAR stable-isotope course that Hannah Marchant and Marcus Elvert taught, I know the ^{15}N isotope-pairing framework — how the $^{28}N_2$: $^{30}N_2$ distribution separates denitrification from anammox, the exetainer time series with its helium purge and mercuric-chloride kill, rates by regression over T0–T4, the pool-dilution and bottle-effect corrections, membrane-inlet mass spectrometry — and my group’s design exercise there was a $^{15}N/^{13}C$ dual-label incubation read out by NanoSIMS. What I have not done is run one on a real sample, and I would rather say so than dress lipid chemistry up as molecular ecology.

So my first experiment is already concrete: Baltic aggregates held in suspension, the oxygen of the water driven between saturation and hypoxia by mass-flow-controlled gas mixing — falling fast, falling slowly, held steady as controls — ^{15}N -nitrate added, N_2 read by isotope pairing on the MIMS, N_2O by gas chromatography, and the carbon arm from my own toolbox beside it. Sequencing-based community characterisation I would be learning from the ground up, and that is the direction I want: further into microbiology, into a biogeochemistry that is designed, incubated and measured rather than inferred — applied science on a real coastal problem. IOW is where I want to do it: one of the few places where a flume in which oxygen can actually be driven, the landers, the stable-isotope and nanoSIMS facilities, the physics department that would tell me what the water is doing, and a vessel with the draught for a genuinely shallow coast all sit in one house, beside the Baltic’s long observational record. I saw your lander at Fabian Lange’s colloquium and have wanted to work with one since; the position asks the student to help organise the field campaigns, and having co-organised a two-week campaign for more than twenty students in the Argentinian Puna, that is work I would take on gladly. Enrolling at Rostock — the oldest university in the Baltic Sea region, and the place where you took your own doctorate, Dr Henkel — is not a formality to me.

I defended my M.Sc. thesis on 23 March 2026 and am free to begin from 1 November 2026. HypoWaves is interdisciplinary by design, and that is exactly why I want it: reading the two of you side by side is what decided me, because the transport physics and the redox microbiology have not yet been joined on a particle — the junction a working group called Physical Microbiology exists to make — and this is the project where that happens.

Yours sincerely,

Osman Can Kandemiroglu

LITERATURE CITED.

Lange, F., Ahmerkamp, S., et al. (2026) Algal glycans promote microbial turnover of organic matter through stabilization of subtidal permeable sandy sediments. *Limnol. Oceanogr.* 71(3), e70350, doi:10.1002/lno.70350 · Voss, M., et al. (2025) Coastal zones of the Baltic Sea in the Anthropocene: current state and the impact of climate change. *Estuar. Coast. Shelf Sci.* 326, 109504, doi:10.1016/j.ecss.2025.109504 · Wan, X. S., et al. (2023) Particle-associated denitrification is the primary source of N_2O in oxic coastal waters. *Nat. Commun.* 14, 8280, doi:10.1038/s41467-023-43997-3 · Marchant, H. K., Ahmerkamp, S., et al. (2018) Metabolic specialization of denitrifiers in permeable sediments controls N_2O emissions. *Environ. Microbiol.* 20(12), 4486–4502, doi:10.1111/1462-2920.14385 · Henkel, J. V., et al. (2022) Biological manganese-dependent sulfide oxidation impacts elemental gradients in redox-stratified systems: indications from the Black Sea water column. *ISME J.* 16, 1523–1533, doi:10.1038/s41396-022-01200-3